CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

LUIS C NUNES & SONS DAIRY #2

10268 S HIGHWAY 33

GUSTINE CA 95322

NOTES:

PREPARED BY:

CARDOSO AG SERVICES
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Luis C Nunes & Sons Dairy #2

Physical address of dairy:

Number and Street	City	County	Zip Code
10268 S Highway 33	Gustine	Merced	95322

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X070-X023-XXXX X070-X070-X037-XXXX

B. OPERATORS

Nunes, Louie

Operator name: Nunes, Louie

Telephone no.: Landline: ; Cellular: (209) 652-7647

Mailing Address Number and Street	City	State	Zip Code
735 Ashley CT	Gustine	CA	95322

This operator is responsible for paying permit fees.

C. OWNERS

Nunes, Louie

Legal owner name: Nunes, Louie

Telephone no.: Landline: ; Cellular: (209) 652-7647

Mailing Address Number and Street	City	State	Zip Code
735 Ashley CT	Gustine	CA	95322

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	415	240	230	180
Number under roof	545	80	0	0	0	0
Maximum number	545	80	415	240	230	180
Average number	545	80	415	240	230	180
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 74 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 23,650.07 *tons per reporting period*

Total nitrogen from manure: 283,837.78 *lbs per reporting period*

After ammonia losses (30% loss applied): 198,686.45 *lbs per reporting period*

Total phosphorus from manure: 45,487.03 *lbs per reporting period*

Total potassium from manure: 105,804.79 *lbs per reporting period*

Total salt from manure: 275,009.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 26,989,607 *gallons*

Total nitrogen generated: 135,443.81 *lbs*

Total phosphorus generated: 21,210.11 *lbs*

Total potassium generated: 182,968.48 *lbs*

Total salt generated: 863,251.28 *lbs*

	26,989,607	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	26,989,607	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
CCID Canal	Surface water
Domestic Well	Ground water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	Separator solids *Solid Manure Imported from Dairy #1*	1,575.00 *ton*	Dry-weight	49.6	21,300.00	5,500.00	19,700.00		0.00
10/18/2023	Separator solids *Solid Manure Imported from Dairy #1*	1,300.00 *ton*	Dry-weight	73.0	29,700.00	9,900.00	42,000.00		0.00

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	54,665.28	15,681.60	60,759.72	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	54,665.28	15,681.60	60,759.72	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Barnett	41	41	1	manure	X070-X270-X001-XXXX
Dairy 2 (A)	40	40	2	process wastewater	X070-X070-X038-XXXX
Dairy 2 (B & C)	119	105	2	process wastewater	X070-X070-X037-XXXX
Fantazia N	38	38	2	manure	X070-X060-X022-XXXX
Fantazia S	40	40	1	manure	X070-X060-X022-XXXX
Mid Cal	77	77	2	manure	X070-X100-X011-XXXX
Santa Nella	80	80	1	manure	X070-X230-X041-XXXX
Seamas North	50	50	2	process wastewater	X070-X070-X004-XXXX
Seamas South	25	25	2	process wastewater	X070-X070-X014-XXXX X070-X070-X026-XXXX X070-X070-X027-XXXX
Terry	40	40	1	manure	X070-X060-X024-XXXX
Totals for areas that were used for application	550	536	16		
Totals for areas that were not used for application					
Land application area totals	550	536	16		

B. CROPS AND HARVESTS

Barnett

Field name: Barnett

10/16/2022: Oats, hay

Crop: Oats, hay Acres planted: 41 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/12/2023	164.00 *ton*	Dry-weight		10.0	22,400.00	3,600.00	32,400.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.00	161.28	25.92	233.28	885.60

Dairy 2 (A)

Dairy 2 (A)

Field name: Dairy 2 (A)

10/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/13/2023	720.00 *ton*	Dry-weight		74.2	16,600.00	3,800.00	27,700.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	154.18	35.29	257.28	1,123.85

04/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 04/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	1,280.00 *ton*	Dry-weight		62.4	12,700.00	2,200.00	13,600.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.00	305.61	52.94	327.27	1,275.39

Dairy 2 (B & C)

Field name: Dairy 2 (B & C)

10/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 105 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	1,890.00 *ton*	Dry-weight		65.9	18,400.00	3,200.00	20,100.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	225.88	39.28	246.75	920.70

Dairy 2 (B & C)

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 105 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	3,360.00 *ton*	Dry-weight		62.2	12,100.00	2,400.00	11,000.00		5.45

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.00	292.72	58.06	266.11	1,318.46

Fantazia N

Field name: Fantazia N

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	684.00 *ton*	Dry-weight		69.2	18,200.00	3,100.00	21,600.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	201.80	34.37	239.50	831.60

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	1,216.00 *ton*	Dry-weight		68.0	11,000.00	2,700.00	13,200.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.00	225.28	55.30	270.34	1,269.76

Fantazia S

Field name: Fantazia S

11/20/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 40 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	320.00 *ton*	Dry-weight		10.3	33,000.00	3,100.00	31,100.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.00	473.62	44.49	446.35	1,420.85

Mid Cal

Field name: Mid Cal

10/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 77 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/12/2023	1,386.00 *ton*	Dry-weight		76.0	12,800.00	2,100.00	18,600.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	110.59	18.14	160.70	794.88

04/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 77 Plant date: 04/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	2,464.00 *ton*	Dry-weight		59.8	12,600.00	1,700.00	9,900.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.00	324.17	43.74	254.71	1,286.40

Santa Nella

Field name: Santa Nella

10/15/2022: Oats, hay

Crop: Oats, hay Acres planted: 80 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/12/2023	320.00 *ton*	Dry-weight		10.0	22,400.00	3,600.00	32,400.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.00	161.28	25.92	233.28	885.60

Seamas North

Field name: Seamas North

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/03/2023	900.00 *ton*	Dry-weight		71.3	13,900.00	3,600.00	23,700.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	143.61	37.20	244.87	1,157.18

04/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 04/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	1,600.00 *ton*	Dry-weight		63.1	13,200.00	3,000.00	12,400.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.00	311.73	70.85	292.84	1,369.73

Seamas South

Field name: Seamas South

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/03/2023	450.00 *ton*	Dry-weight		71.3	13,900.00	3,600.00	23,700.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.00	143.61	37.20	244.87	1,157.18

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	800.00 *ton*	Dry-weight		63.1	13,200.00	3,000.00	12,400.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.00	311.73	70.85	292.84	1,369.73

Terry

Field name: Terry

10/12/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 40 Plant date: 10/12/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/28/2023	320.00 *ton*	Dry-weight		12.6	32,100.00	3,100.00	34,100.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.00	448.89	43.35	476.85	1,412.38

NUTRIENT BUDGET

A. LAND APPLICATIONS

Barnett - 10/16/2022: Oats, hay

Field name: Barnett

Crop: Oats, hay — Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Scraped material	197.43	39.49	277.08	0.00	350.00 *ton*
Application event totals		197.43	39.49	277.08	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.34	0.00	0.00	383.35	9,417,263.00 *gal*
Application event totals		1.34	0.00	0.00	383.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.01	0.00	0.00	289.66	7,115,691.00 *gal*
Application event totals		1.01	0.00	0.00	289.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.00	0.00	0.00	285.58	7,015,530.00 *gal*
Application event totals		1.00	0.00	0.00	285.58	

Dairy 2 (A) - 10/15/2022: Oats, silage-soft dough

Field name: Dairy 2 (A)

Crop: Oats, silage-soft dough — Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Dairy 2 (A) - 10/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	53.23	9.28	71.65	422.47	540,000.00 *gal*
CCID Canal	Surface water	0.56	0.00	0.00	159.27	3,817,240.00 *gal*
Application event totals		53.79	9.28	71.65	581.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	56.78	9.90	76.43	450.63	576,000.00 *gal*
CCID Canal	Surface water	0.47	0.00	0.00	132.90	3,185,192.00 *gal*
Application event totals		57.24	9.90	76.43	583.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.17	17.66	83.73	450.63	432,000.00 *gal*
CCID Canal	Surface water	0.47	0.00	0.00	133.60	3,201,825.00 *gal*
Application event totals		91.64	17.66	83.73	584.23	

Dairy 2 (A) - 04/28/2023: Corn, silage

Field name: Dairy 2 (A)

Crop: Corn, silage

Plant date: 04/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.49	15.84	99.46	437.43	462,857.00 *gal*
CCID Canal	Surface water	1.34	0.00	0.00	383.35	9,187,573.00 *gal*
Application event totals		99.84	15.84	99.46	820.78	

Dairy 2 (A) - 04/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.99	14.04	167.83	710.99	960,000.00 *gal*
CCID Canal	Surface water	1.01	0.00	0.00	289.66	6,942,138.00 *gal*
Application event totals		105.00	14.04	167.83	1,000.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.00	0.00	0.00	285.58	6,844,420.00 *gal*
Application event totals		1.00	0.00	0.00	285.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.61	12.50	149.48	633.23	855,000.00 *gal*
CCID Canal	Surface water	1.08	0.00	0.00	307.97	7,380,829.00 *gal*
Application event totals		93.69	12.50	149.48	941.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.49	13.16	157.34	666.56	900,000.00 *gal*
CCID Canal	Surface water	1.27	0.00	0.00	364.01	8,723,932.00 *gal*
Application event totals		98.76	13.16	157.34	1,030.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.29	0.00	0.00	368.52	8,832,046.00 *gal*
Application event totals		1.29	0.00	0.00	368.52	

Dairy 2 (A) - 04/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.19	0.00	0.00	340.15	8,152,178.00 *gal*
Application event totals		1.19	0.00	0.00	340.15	

Dairy 2 (B & C) - 10/15/2022: Oats, silage-soft dough

Field name: Dairy 2 (B & C)

Crop: Oats, silage-soft dough

Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.11	14.15	109.18	643.76	2,160,000.00 *gal*
CCID Canal	Surface water	0.56	0.00	0.00	159.27	10,020,256.00 *gal*
Application event totals		81.67	14.15	109.18	803.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.49	14.73	113.73	670.58	2,250,000.00 *gal*
CCID Canal	Surface water	0.47	0.00	0.00	132.90	8,361,128.00 *gal*
Application event totals		84.96	14.73	113.73	803.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.30	19.63	93.03	500.70	1,260,000.00 *gal*
CCID Canal	Surface water	0.47	0.00	0.00	133.60	8,404,789.00 *gal*
Application event totals		101.77	19.63	93.03	634.30	

Dairy 2 (B & C) - 05/28/2023: Corn, silage

Dairy 2 (B & C) - 05/28/2023: Corn, silage

Field name: Dairy 2 (B & C)

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.49	15.84	99.46	437.43	1,215,000.00 *gal*
CCID Canal	Surface water	1.34	0.00	0.00	383.35	24,117,380.00 *gal*
Application event totals		99.84	15.84	99.46	820.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.99	14.04	167.83	710.99	2,520,000.00 *gal*
CCID Canal	Surface water	1.01	0.00	0.00	289.66	18,223,112.00 *gal*
Application event totals		105.00	14.04	167.83	1,000.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.00	0.00	0.00	285.58	17,966,602.00 *gal*
Application event totals		1.00	0.00	0.00	285.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.84	12.54	149.85	634.82	2,250,000.00 *gal*
CCID Canal	Surface water	1.08	0.00	0.00	307.97	19,374,677.00 *gal*
Application event totals		93.92	12.54	149.85	942.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.56	13.04	155.85	660.21	2,340,000.00 *gal*
CCID Canal	Surface water	1.27	0.00	0.00	364.01	22,900,322.00 *gal*
Application event totals		97.83	13.04	155.85	1,024.21	

Dairy 2 (B & C) - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.29	0.00	0.00	368.52	23,184,121.00 *gal*
Application event totals		1.29	0.00	0.00	368.52	

09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.19	0.00	0.00	340.15	21,399,467.00 *gal*
Application event totals		1.19	0.00	0.00	340.15	

Fantazia N - 10/16/2022: Oats, silage-soft dough

Field name: Fantazia N

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Dairy #1	Separator solids	253.23	84.41	358.11	0.00	600.00 *ton*
Application event totals		253.23	84.41	358.11	0.00	

10/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.56	0.00	0.00	159.27	3,626,378.00 *gal*
Application event totals		0.56	0.00	0.00	159.27	

11/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.47	0.00	0.00	132.90	3,025,932.00 *gal*
Application event totals		0.47	0.00	0.00	132.90	

Fantazia N - 10/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.47	0.00	0.00	133.60	3,041,733.00 *gal*
Application event totals		0.47	0.00	0.00	133.60	

Fantazia N - 05/28/2023: Corn, silage

Field name: Fantazia N

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Dairy #1	Separator solids	296.63	76.59	274.35	0.00	525.00 *ton*
Application event totals		296.63	76.59	274.35	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.34	0.00	0.00	383.35	8,728,195.00 *gal*
Application event totals		1.34	0.00	0.00	383.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.01	0.00	0.00	289.66	6,595,031.00 *gal*
Application event totals		1.01	0.00	0.00	289.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.00	0.00	0.00	285.58	6,502,199.00 *gal*
Application event totals		1.00	0.00	0.00	285.58	

Fantazia N - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.08	0.00	0.00	307.97	7,011,788.00 *gal*
Application event totals		1.08	0.00	0.00	307.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.27	0.00	0.00	364.01	8,287,736.00 *gal*
Application event totals		1.27	0.00	0.00	364.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.29	0.00	0.00	368.52	8,390,444.00 *gal*
Application event totals		1.29	0.00	0.00	368.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.19	0.00	0.00	340.15	7,744,569.00 *gal*
Application event totals		1.19	0.00	0.00	340.15	

Fantazia S - 11/20/2022: Alfalfa, hay

Field name: Fantazia S

Crop: Alfalfa, hay

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.76	0.00	0.00	217.76	5,218,834.00 *gal*
Application event totals		0.76	0.00	0.00	217.76	

Fantazia S - 11/20/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.31	0.00	0.00	373.59	8,953,578.00 *gal*
Application event totals		1.31	0.00	0.00	373.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.31	0.00	0.00	375.04	8,988,267.00 *gal*
Application event totals		1.31	0.00	0.00	375.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.13	0.00	0.00	322.01	7,717,531.00 *gal*
Application event totals		1.13	0.00	0.00	322.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.10	0.00	0.00	313.33	7,509,393.00 *gal*
Application event totals		1.10	0.00	0.00	313.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.29	0.00	0.00	369.32	8,851,334.00 *gal*
Application event totals		1.29	0.00	0.00	369.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.00	0.00	0.00	286.44	6,864,896.00 *gal*
Application event totals		1.00	0.00	0.00	286.44	

Fantazia S - 11/20/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.89	0.00	0.00	254.14	6,090,770.00 *gal*
Application event totals		0.89	0.00	0.00	254.14	

10/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Dairy #1	Separator solids	280.67	93.56	396.90	0.00	700.00 *ton*
Application event totals		280.67	93.56	396.90	0.00	

Mid Cal - 10/15/2022: Oats, silage-soft dough

Field name: Mid Cal

Crop: Oats, silage-soft dough

Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Scraped material	145.68	29.14	204.44	0.00	485.00 *ton*
Application event totals		145.68	29.14	204.44	0.00	

10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.56	0.00	0.00	159.27	7,348,187.00 *gal*
Application event totals		0.56	0.00	0.00	159.27	

11/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.47	0.00	0.00	132.90	6,131,494.00 *gal*
Application event totals		0.47	0.00	0.00	132.90	

Mid Cal - 10/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.47	0.00	0.00	133.60	6,163,512.00 *gal*
Application event totals		0.47	0.00	0.00	133.60	

Mid Cal - 04/28/2023: Corn, silage

Field name: Mid Cal

Crop: Corn, silage

Plant date: 04/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure Imported from Dairy #1	Separator solids	292.78	75.60	270.79	0.00	1,050.00 *ton*
Application event totals		292.78	75.60	270.79	0.00	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.34	0.00	0.00	383.35	17,686,078.00 *gal*
Application event totals		1.34	0.00	0.00	383.35	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.01	0.00	0.00	289.66	13,363,615.00 *gal*
Application event totals		1.01	0.00	0.00	289.66	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.00	0.00	0.00	285.58	13,175,508.00 *gal*
Application event totals		1.00	0.00	0.00	285.58	

Mid Cal - 04/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.08	0.00	0.00	307.97	14,208,096.00 *gal*
Application event totals		1.08	0.00	0.00	307.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.27	0.00	0.00	364.01	16,793,570.00 *gal*
Application event totals		1.27	0.00	0.00	364.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.29	0.00	0.00	368.52	17,001,688.00 *gal*
Application event totals		1.29	0.00	0.00	368.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.19	0.00	0.00	340.15	15,692,943.00 *gal*
Application event totals		1.19	0.00	0.00	340.15	

Santa Nella - 10/15/2022: Oats, hay

Field name: Santa Nella

Crop: Oats, hay

Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Scraped material	195.14	39.03	273.86	0.00	675.00 *ton*
Application event totals		195.14	39.03	273.86	0.00	

Seamas North - 10/16/2022: Oats, silage-soft dough

Field name: Seamas North

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.56	0.00	0.00	159.27	4,771,550.00 *gal*
Application event totals		0.56	0.00	0.00	159.27	

11/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.07	13.62	105.09	619.62	990,000.00 *gal*
CCID Canal	Surface water	0.47	0.00	0.00	132.90	3,981,490.00 *gal*
Application event totals		78.54	13.62	105.09	752.52	

03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.77	19.14	90.70	488.18	585,000.00 *gal*
CCID Canal	Surface water	0.47	0.00	0.00	133.60	4,002,281.00 *gal*
Application event totals		99.24	19.14	90.70	621.78	

Seamas North - 04/28/2023: Corn, silage

Field name: Seamas North

Crop: Corn, silage

Plant date: 04/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.59	16.01	100.57	442.29	585,000.00 *gal*
CCID Canal	Surface water	1.34	0.00	0.00	383.35	11,484,467.00 *gal*
Application event totals		100.93	16.01	100.57	825.64	

Seamas North - 04/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.01	0.00	0.00	289.66	8,677,672.00 *gal*
Application event totals		1.01	0.00	0.00	289.66	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.49	13.16	157.34	666.56	1,125,000.00 *gal*
CCID Canal	Surface water	1.00	0.00	0.00	285.58	8,555,525.00 *gal*
Application event totals		98.49	13.16	157.34	952.14	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.59	12.64	151.05	639.89	1,080,000.00 *gal*
CCID Canal	Surface water	1.08	0.00	0.00	307.97	9,226,037.00 *gal*
Application event totals		94.66	12.64	151.05	947.86	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.27	0.00	0.00	364.01	10,904,915.00 *gal*
Application event totals		1.27	0.00	0.00	364.01	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.44	13.43	160.49	679.89	1,147,500.00 *gal*
CCID Canal	Surface water	1.29	0.00	0.00	368.52	11,040,057.00 *gal*
Application event totals		100.73	13.43	160.49	1,048.41	

Seamas North - 04/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.19	0.00	0.00	340.15	10,190,222.00 *gal*
Application event totals		1.19	0.00	0.00	340.15	

Seamas South - 10/16/2022: Oats, silage-soft dough

Field name: Seamas South

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.56	0.00	0.00	159.27	2,385,775.00 *gal*
Application event totals		0.56	0.00	0.00	159.27	

11/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.07	13.62	105.09	619.62	495,000.00 *gal*
CCID Canal	Surface water	0.47	0.00	0.00	132.90	1,990,745.00 *gal*
Application event totals		78.54	13.62	105.09	752.52	

03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.77	19.14	90.70	488.18	292,500.00 *gal*
CCID Canal	Surface water	0.47	0.00	0.00	133.60	2,001,140.00 *gal*
Application event totals		99.24	19.14	90.70	621.78	

Seamas South - 05/28/2023: Corn, silage

Seamas South - 05/28/2023: Corn, silage

Field name: Seamas South

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.59	16.01	100.57	442.29	292,500.00 *gal*
CCID Canal	Surface water	1.34	0.00	0.00	383.35	5,742,233.00 *gal*
Application event totals		100.93	16.01	100.57	825.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.01	0.00	0.00	289.66	4,338,836.00 *gal*
Application event totals		1.01	0.00	0.00	289.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.49	13.16	157.34	666.56	562,500.00 *gal*
CCID Canal	Surface water	1.00	0.00	0.00	285.58	4,277,762.00 *gal*
Application event totals		98.49	13.16	157.34	952.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.59	12.64	151.05	639.89	540,000.00 *gal*
CCID Canal	Surface water	1.08	0.00	0.00	307.97	4,613,018.00 *gal*
Application event totals		94.66	12.64	151.05	947.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.27	0.00	0.00	364.01	5,452,458.00 *gal*
Application event totals		1.27	0.00	0.00	364.01	

Seamas South - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.44	13.43	160.49	679.89	573,750.00 *gal*
CCID Canal	Surface water	1.29	0.00	0.00	368.52	5,520,029.00 *gal*
Application event totals		100.73	13.43	160.49	1,048.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.19	0.00	0.00	340.15	5,095,111.00 *gal*
Application event totals		1.19	0.00	0.00	340.15	

Terry - 10/12/2021: Alfalfa, hay

Field name: Terry

Crop: Alfalfa, hay

Plant date: 10/12/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.76	0.00	0.00	217.76	5,218,834.00 *gal*
Application event totals		0.76	0.00	0.00	217.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.31	0.00	0.00	373.59	8,953,578.00 *gal*
Application event totals		1.31	0.00	0.00	373.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.31	0.00	0.00	375.04	8,988,267.00 *gal*
Application event totals		1.31	0.00	0.00	375.04	

Terry - 10/12/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.13	0.00	0.00	322.01	7,717,531.00 *gal*
Application event totals		1.13	0.00	0.00	322.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.10	0.00	0.00	313.33	7,509,393.00 *gal*
Application event totals		1.10	0.00	0.00	313.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.29	0.00	0.00	369.32	8,851,334.00 *gal*
Application event totals		1.29	0.00	0.00	369.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	1.00	0.00	0.00	286.44	6,864,896.00 *gal*
Application event totals		1.00	0.00	0.00	286.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID Canal	Surface water	0.89	0.00	0.00	254.14	6,090,770.00 *gal*
Application event totals		0.89	0.00	0.00	254.14	

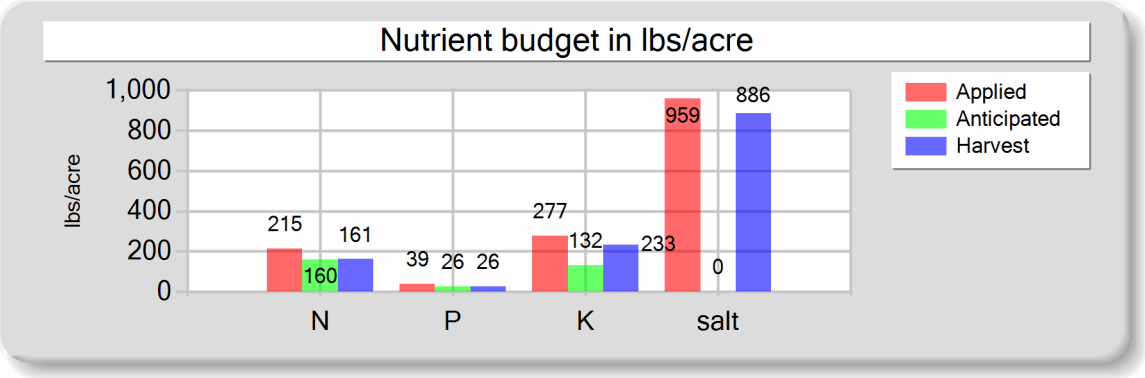
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	295.06	141.48	549.00	0.00	350.00 *ton*
Application event totals		295.06	141.48	549.00	0.00	

B. NUTRIENT BUDGET

Barnett - 10/16/2022: Oats, hay

Field name: Barnett Crop: Oats, hay Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	197.43	39.49	277.08	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	3.36	0.00	0.00	958.60
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	214.79	39.49	277.08	958.60
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	161.28	25.92	233.28	885.60
Nutrient balance	53.51	13.57	43.80	73.00
Applied to removed ratio	1.33	1.52	1.19	1.08

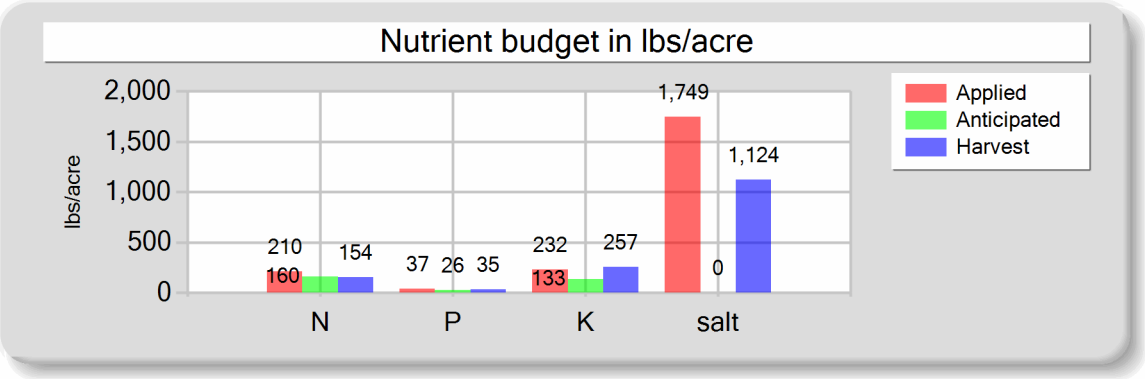
Fresh water applied	
23,548,484.00	*gallons*
867.21	*acre-inches*
21.15	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 2 (A) - 10/15/2022: Oats, silage-soft dough

Field name: Dairy 2 (A) Crop: Oats, silage-soft dough Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.18	36.85	231.80	1,323.73
Fresh water	1.49	0.00	0.00	425.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	209.67	36.85	231.80	1,749.50
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	154.18	35.29	257.28	1,123.85
Nutrient balance	55.49	1.56	-25.47	625.65
Applied to removed ratio	1.36	1.04	0.90	1.56

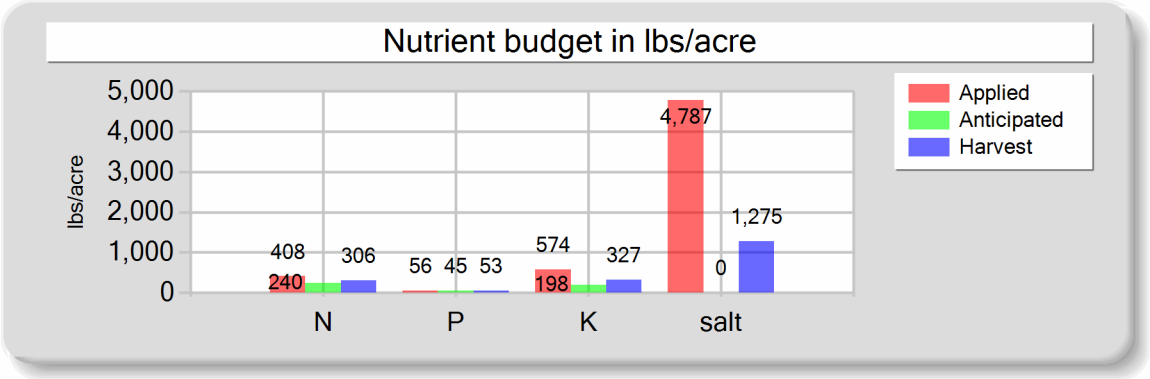
Fresh water applied	
10,204,257.00	*gallons*
375.79	*acre-inches*
9.39	*inches/acre*

Process wastewater applied	
1,548,000.00	*gallons*
57.01	*acre-inches*
1.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 2 (A) - 04/28/2023: Corn, silage

Field name: Dairy 2 (A) Crop: Corn, silage Plant date: 04/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	392.58	55.54	574.12	2,448.21
Fresh water	8.19	0.00	0.00	2,339.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	407.77	55.54	574.12	4,787.45
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	305.61	52.94	327.27	1,275.39
Nutrient balance	102.15	2.60	246.85	3,512.05
Applied to removed ratio	1.33	1.05	1.75	3.75

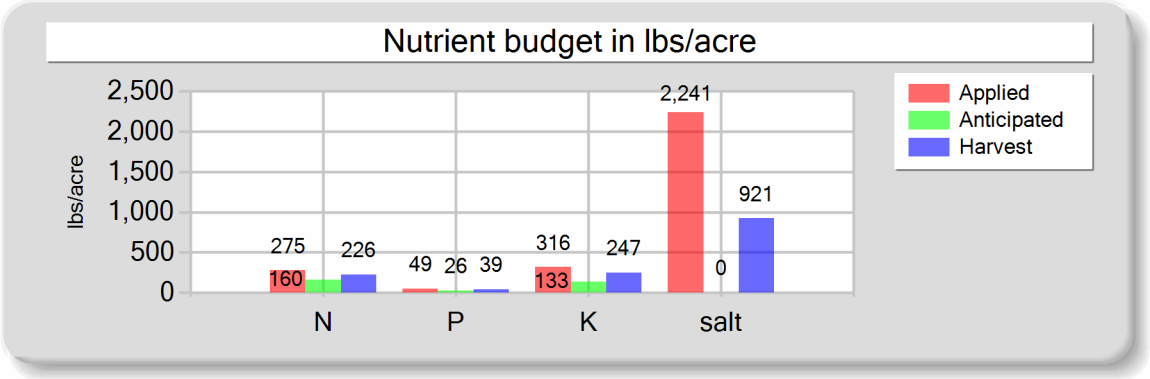
Fresh water applied	
56,063,116.00	*gallons*
2,064.61	*acre-inches*
51.62	*inches/acre*

Process wastewater applied	
3,177,857.00	*gallons*
117.03	*acre-inches*
2.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 2 (B & C) - 10/15/2022: Oats, silage-soft dough

Field name: Dairy 2 (B & C) Crop: Oats, silage-soft dough Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	266.91	48.51	315.94	1,815.04
Fresh water	1.49	0.00	0.00	425.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	275.40	48.51	315.94	2,240.81
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	225.88	39.28	246.75	920.70
Nutrient balance	49.52	9.22	69.19	1,320.11
Applied to removed ratio	1.22	1.23	1.28	2.43

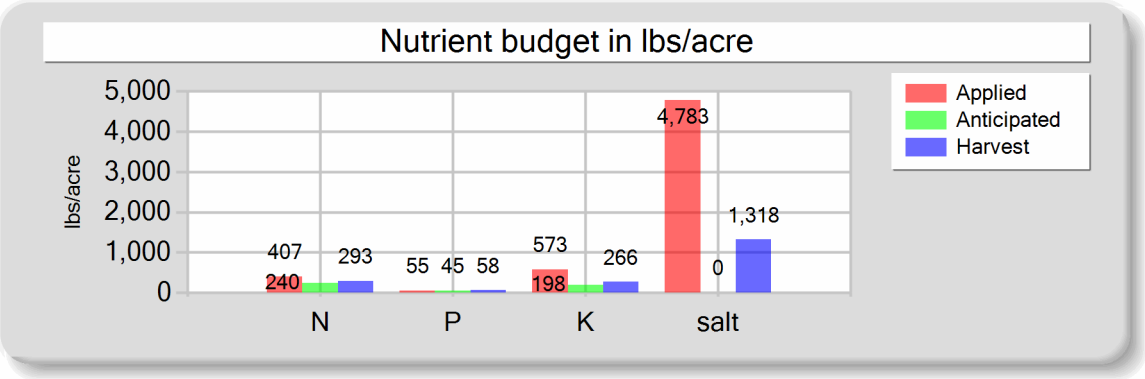
Fresh water applied	
26,786,173.00	*gallons*
986.44	*acre-inches*
9.39	*inches/acre*

Process wastewater applied	
5,670,000.00	*gallons*
208.81	*acre-inches*
1.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 2 (B & C) - 05/28/2023: Corn, silage

Field name: Dairy 2 (B & C) Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	391.88	55.45	572.99	2,443.45
Fresh water	8.19	0.00	0.00	2,339.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	407.07	55.45	572.99	4,782.69
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	292.72	58.06	266.11	1,318.46
Nutrient balance	114.35	-2.61	306.88	3,464.22
Applied to removed ratio	1.39	0.96	2.15	3.63

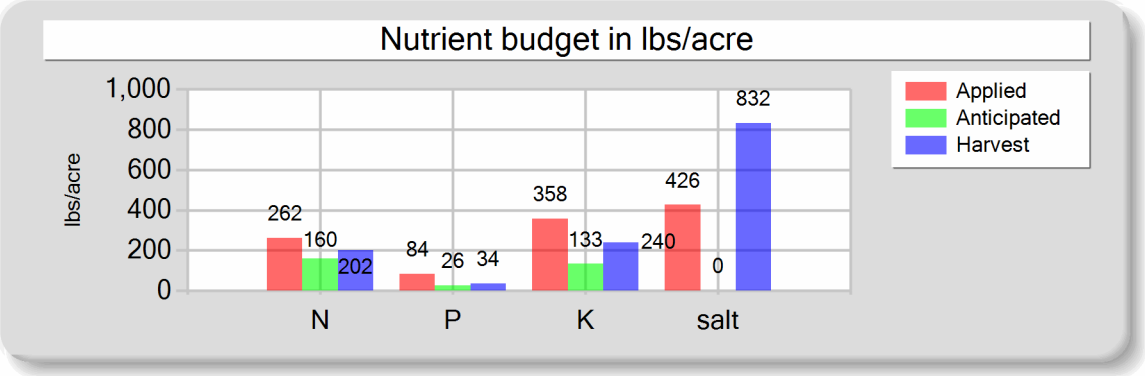
Fresh water applied	
147,165,681.00	*gallons*
5,419.61	*acre-inches*
51.62	*inches/acre*

Process wastewater applied	
8,325,000.00	*gallons*
306.58	*acre-inches*
2.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fantazia N - 10/16/2022: Oats, silage-soft dough

Field name: Fantazia N Crop: Oats, silage-soft dough Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	253.23	84.41	358.11	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.49	0.00	0.00	425.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.72	84.41	358.11	425.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	201.80	34.37	239.50	831.60
Nutrient balance	59.92	50.04	118.60	-405.83
Applied to removed ratio	1.30	2.46	1.50	0.51

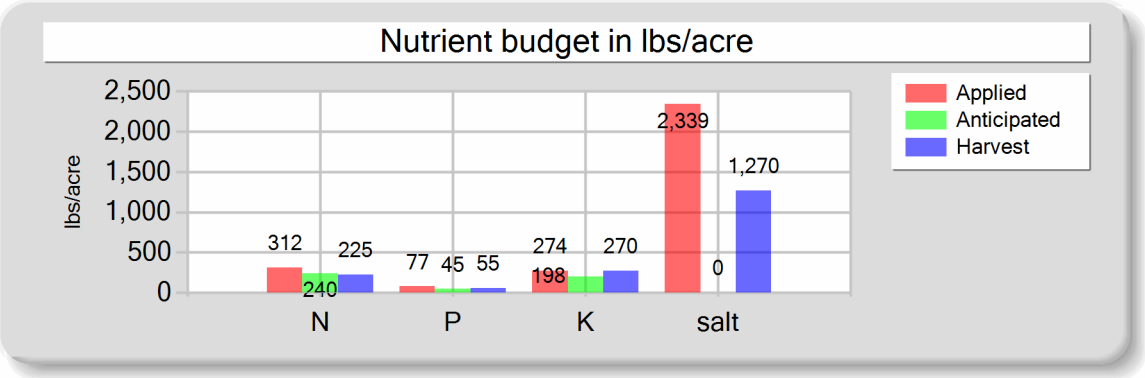
Fresh water applied	
9,694,043.00	*gallons*
357.00	*acre-inches*
9.39	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fantazia N - 05/28/2023: Corn, silage

Field name: Fantazia N Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	296.63	76.59	274.35	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.19	0.00	0.00	2,339.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	311.82	76.59	274.35	2,339.23
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	225.28	55.30	270.34	1,269.76
Nutrient balance	86.54	21.30	4.01	1,069.47
Applied to removed ratio	1.38	1.39	1.01	1.84

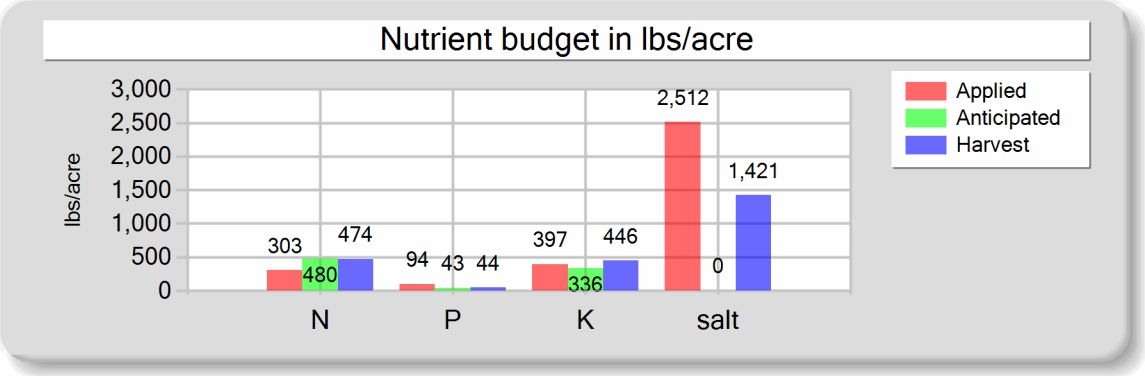
Fresh water applied	
53,259,962.00	*gallons*
1,961.38	*acre-inches*
51.62	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fantazia S - 11/20/2022: Alfalfa, hay

Field name: Fantazia S Crop: Alfalfa, hay Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	280.67	93.56	396.90	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.79	0.00	0.00	2,511.62
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	303.46	93.56	396.90	2,511.62
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	473.62	44.49	446.35	1,420.85
Nutrient balance	-170.16	49.06	-49.45	1,090.77
Applied to removed ratio	0.64	2.10	0.89	1.77

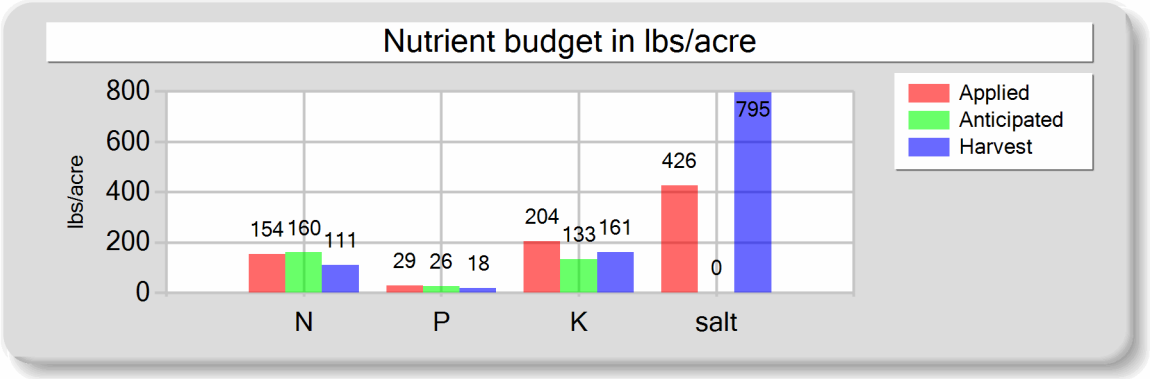
Fresh water applied	
60,194,603.00	*gallons*
2,216.76	*acre-inches*
55.42	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Mid Cal - 10/15/2022: Oats, silage-soft dough

Field name: Mid Cal | Crop: Oats, silage-soft dough | Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	145.68	29.14	204.44	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.49	0.00	0.00	425.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.17	29.14	204.44	425.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	110.59	18.14	160.70	794.88
Nutrient balance	43.57	10.99	43.74	-369.11
Applied to removed ratio	1.39	1.61	1.27	0.54

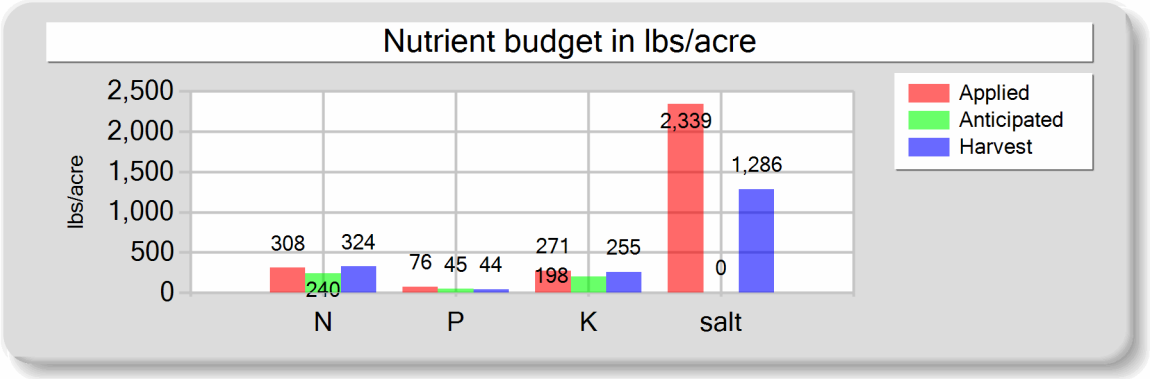
Fresh water applied	
19,643,193.00	*gallons*
723.39	*acre-inches*
9.39	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Mid Cal - 04/28/2023: Corn, silage

Field name: Mid Cal | Crop: Corn, silage | Plant date: 04/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	292.78	75.60	270.79	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.19	0.00	0.00	2,339.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	307.97	75.60	270.79	2,339.23
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	324.17	43.74	254.71	1,286.40
Nutrient balance	-16.21	31.86	16.08	1,052.83
Applied to removed ratio	0.95	1.73	1.06	1.82

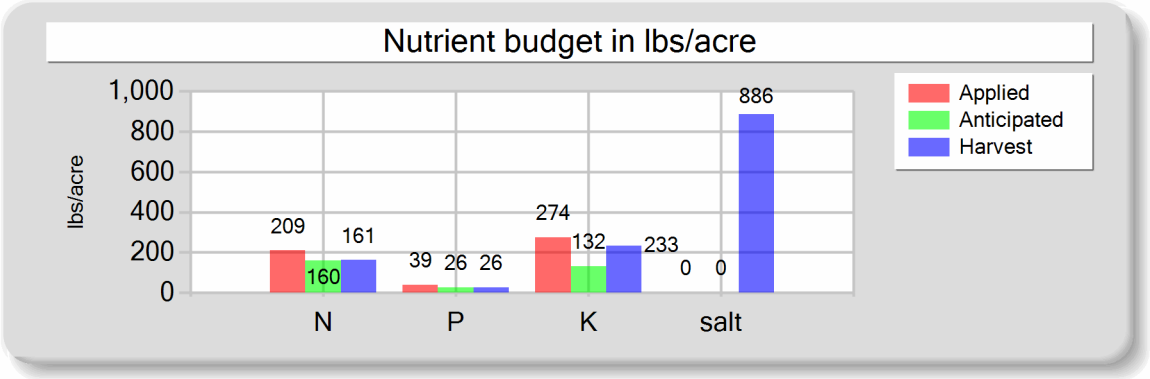
Fresh water applied	
107,921,498.00	*gallons*
3,974.38	*acre-inches*
51.62	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Santa Nella - 10/15/2022: Oats, hay

Field name: Santa Nella Crop: Oats, hay Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	195.14	39.03	273.86	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	209.14	39.03	273.86	0.00
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	161.28	25.92	233.28	885.60
Nutrient balance	47.86	13.11	40.58	-885.60
Applied to removed ratio	1.30	1.51	1.17	0.00

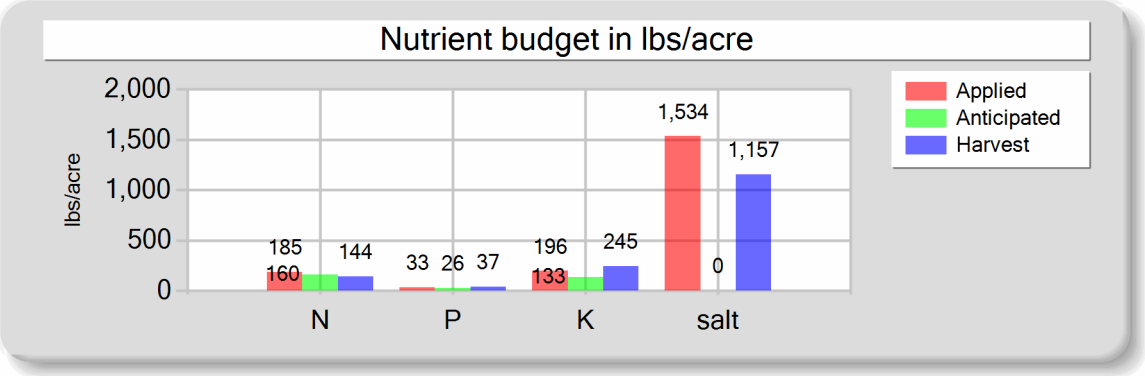
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Seamas North - 10/16/2022: Oats, silage-soft dough

Field name: Seamas North | Crop: Oats, silage-soft dough | Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	176.84	32.75	195.79	1,107.80
Fresh water	1.49	0.00	0.00	425.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.33	32.75	195.79	1,533.57
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	143.61	37.20	244.87	1,157.18
Nutrient balance	41.72	-4.44	-49.08	376.39
Applied to removed ratio	1.29	0.88	0.80	1.33

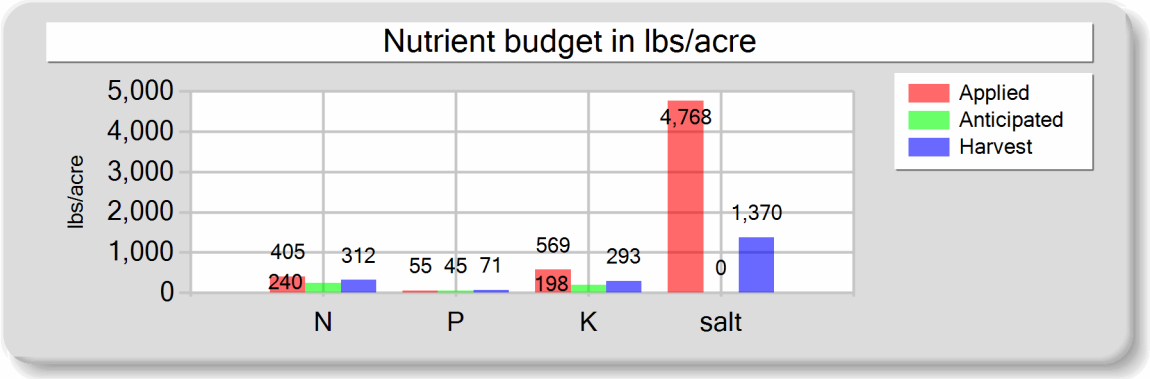
Fresh water applied	
12,755,321.00	*gallons*
469.74	*acre-inches*
9.39	*inches/acre*

Process wastewater applied	
1,575,000.00	*gallons*
58.00	*acre-inches*
1.16	*inches/acre*

Total harvests for the crop	
1	*harvests*

Seamas North - 04/28/2023: Corn, silage

Field name: Seamas North | Crop: Corn, silage | Plant date: 04/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	390.10	55.24	569.45	2,428.63
Fresh water	8.19	0.00	0.00	2,339.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	405.29	55.24	569.45	4,767.87
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	311.73	70.85	292.84	1,369.73
Nutrient balance	93.55	-15.61	276.62	3,398.14
Applied to removed ratio	1.30	0.78	1.94	3.48

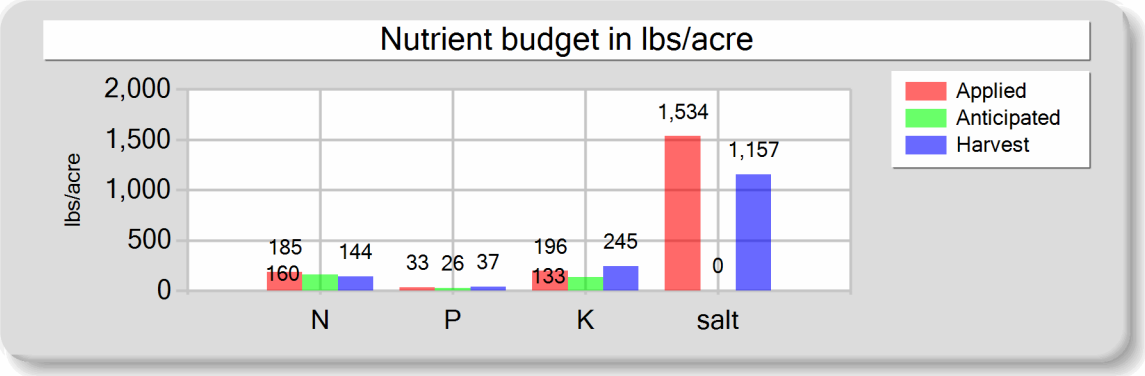
Fresh water applied	
70,078,895.00	*gallons*
2,580.77	*acre-inches*
51.62	*inches/acre*

Process wastewater applied	
3,937,500.00	*gallons*
145.00	*acre-inches*
2.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

Seamas South - 10/16/2022: Oats, silage-soft dough

Field name: Seamas South | Crop: Oats, silage-soft dough | Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	176.84	32.75	195.79	1,107.80
Fresh water	1.49	0.00	0.00	425.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.33	32.75	195.79	1,533.57
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	143.61	37.20	244.87	1,157.18
Nutrient balance	41.72	-4.44	-49.08	376.39
Applied to removed ratio	1.29	0.88	0.80	1.33

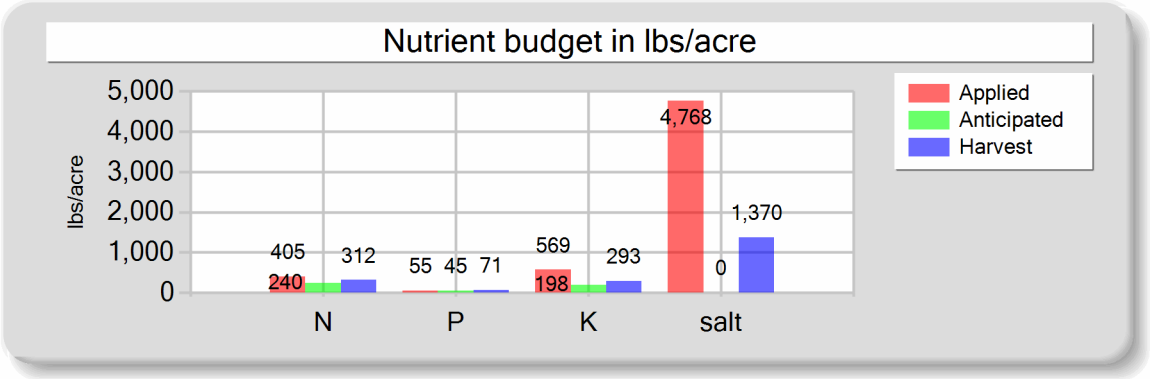
Fresh water applied	
6,377,660.00	*gallons*
234.87	*acre-inches*
9.39	*inches/acre*

Process wastewater applied	
787,500.00	*gallons*
29.00	*acre-inches*
1.16	*inches/acre*

Total harvests for the crop	
1	*harvests*

Seamas South - 05/28/2023: Corn, silage

Field name: Seamas South | Crop: Corn, silage | Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	390.10	55.24	569.45	2,428.63
Fresh water	8.19	0.00	0.00	2,339.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	405.29	55.24	569.45	4,767.87
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	311.73	70.85	292.84	1,369.73
Nutrient balance	93.55	-15.61	276.62	3,398.14
Applied to removed ratio	1.30	0.78	1.94	3.48

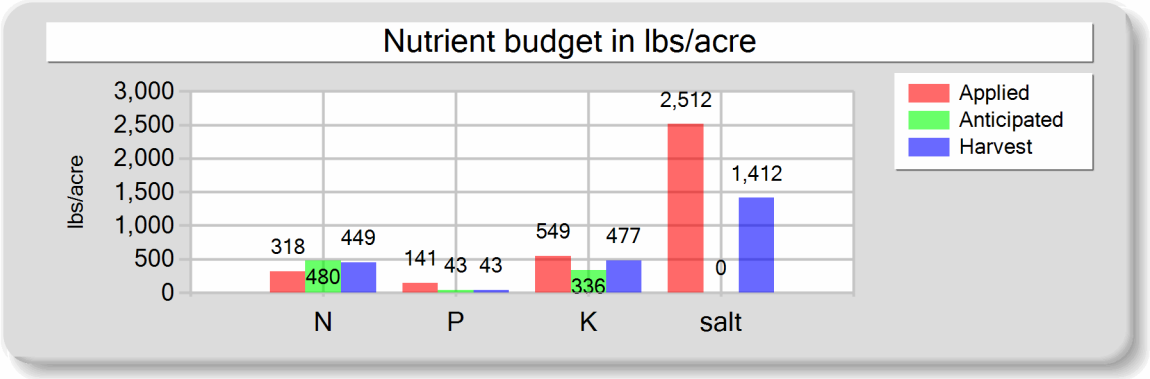
Fresh water applied	
35,039,447.00	*gallons*
1,290.38	*acre-inches*
51.62	*inches/acre*

Process wastewater applied	
1,968,750.00	*gallons*
72.50	*acre-inches*
2.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

Terry - 10/12/2021: Alfalfa, hay

Field name: Terry | Crop: Alfalfa, hay | Plant date: 10/12/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	295.06	141.48	549.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	8.79	0.00	0.00	2,511.62
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	317.85	141.48	549.00	2,511.62
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	448.89	43.35	476.85	1,412.38
Nutrient balance	-131.04	98.13	72.15	1,099.24
Applied to removed ratio	0.71	3.26	1.15	1.78

Fresh water applied	
60,194,603.00	*gallons*
2,216.76	*acre-inches*
55.42	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

SSB Manure - 2022

Sample and source description: SSB Manure - 2022

Sample date: 10/05/2022 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,500.00	5,900.00	41,400.00							0.00
DL	100.00	100.00	100.00							1.00

SSB Manure

Sample and source description: SSB Manure

Sample date: 05/18/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	3,200.00	12,700.00							0.00
DL	0.01	0.01	0.01							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 10/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 30.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,400.00	11,700.00	45,400.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 - 2022

Sample and source description: Lagoon Q4 - 2022

Sample date: 11/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	465.00	336.00	0.00	7.50	82.40	636.00								7,610.00	3,750
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,010.00	484.00	0.00	1.60	196.00	929.00								10,100.00	5,000
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.20

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,020.00	497.00	0.00	0.00	164.00	1,030.00								7,550.00	4,530
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	518.00	383.00	0.00	1.20	70.10	838.00								8,040.00	3,550
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/06/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	590.00	366.00	0.00	0.00	71.90	970.00								9,340.00	4,920
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

CCID Canal

CCID Canal

Sample description: CCID Canal

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.70	0.00	0.70								328.00	200
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW MB

Sample description: DW MB

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.20	0.00	9.20								1,360.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

Irrigation Well

IW

Sample description: IW

Sample date: 08/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.60	0.00	7.60								1,290.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Barnett - 10/16/2022: Oats, hay

Barnett Oat Hay

Sample and source description: Barnett Oat Hay

Sample date: 04/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	3,600.00	32,400.00		12.30
DL	100.00	100.00	10.00		1.00

Dairy 2 (A) - 10/15/2022: Oats, silage-soft dough

Dairy 2 (A) - 10/15/2022: Oats, silage-soft dough

Dairy 2A Oats

Sample and source description: Dairy 2A Oats

Sample date: 04/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,600.00	3,800.00	27,700.00		12.10
DL	0.01	0.01	0.01		1.00

Dairy 2 (A) - 04/28/2023: Corn, silage

Corn Dairy 2A

Sample and source description: Corn Dairy 2A

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	2,200.00	13,600.00		5.30
DL	0.01	0.01	0.01		1.00

Dairy 2 (B & C) - 10/15/2022: Oats, silage-soft dough

Oats Dairy 2-2B

Sample and source description: Oats Dairy 2-2B

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	3,200.00	20,100.00		7.50
DL	0.01	0.01	0.01		1.00

Dairy 2 (B & C) - 05/28/2023: Corn, silage

Corn Dairy 2B

Sample and source description: Corn Dairy 2B

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	2,000.00	10,900.00		5.40
DL	0.01	0.01	0.01		1.00

Corn Dairy 2C

Sample and source description: Corn Dairy 2C

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	2,800.00	11,100.00		5.50
DL	0.01	0.01	0.01		1.00

Fantazia N - 10/16/2022: Oats, silage-soft dough

Oats Fantazia S

Sample and source description: Oats Fantazia S

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	3,100.00	21,600.00		7.50
DL	0.01	0.01	0.01		1.00

Fantazia N - 05/28/2023: Corn, silage

Fantazia N - 05/28/2023: Corn, silage

Fantazia S Corn

Sample and source description: Fantazia S Corn

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,000.00	2,700.00	13,200.00		6.20
DL	0.01	0.01	0.01		1.00

Fantazia S - 11/20/2022: Alfalfa, hay

Fantazia South Alfalfa

Sample and source description: Fantazia South Alfalfa

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,000.00	3,100.00	31,100.00		9.90
DL	100.00	100.00	100.00		1.00

Mid Cal - 10/15/2022: Oats, silage-soft dough

Midcal Oats

Sample and source description: Midcal Oats

Sample date: 04/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,800.00	2,100.00	18,600.00		9.20
DL	0.01	0.01	0.01		1.00

Mid Cal - 04/28/2023: Corn, silage

Corn Mid Cal

Sample and source description: Corn Mid Cal

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	1,700.00	9,900.00		5.00
DL	0.01	0.01	0.01		1.00

Santa Nella - 10/15/2022: Oats, hay

Santa Nella Oats

Sample and source description: Santa Nella Oats

Sample date: 04/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	3,600.00	32,400.00		12.30
DL	0.01	0.01	0.01		1.00

Seamas North - 10/16/2022: Oats, silage-soft dough

Seamas North Oats

Sample and source description: Seamas North Oats

Sample date: 04/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	3,600.00	23,700.00		11.20
DL	0.01	0.01	0.01		1.00

Seamas North - 04/28/2023: Corn, silage

Corn Semas N

Sample and source description: Corn Semas N

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	3,000.00	12,400.00		5.80
DL	0.01	0.01	0.01		1.00

Seamas South - 10/16/2022: Oats, silage-soft dough

Seamas South Oats

Sample and source description: Seamas South Oats

Sample date: 04/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	3,600.00	23,700.00		11.20
DL	100.00	100.00	100.00		1.00

Seamas South - 05/28/2023: Corn, silage

Corn Semas S

Sample and source description: Corn Semas S

Sample date: 10/03/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	3,000.00	12,400.00		5.80
DL	0.01	0.01	0.01		1.00

Terry - 10/12/2021: Alfalfa, hay

Terry Alfalfa

Sample and source description: Terry Alfalfa

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 12.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,100.00	3,100.00	34,100.00		10.10
DL	100.00	100.00	100.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

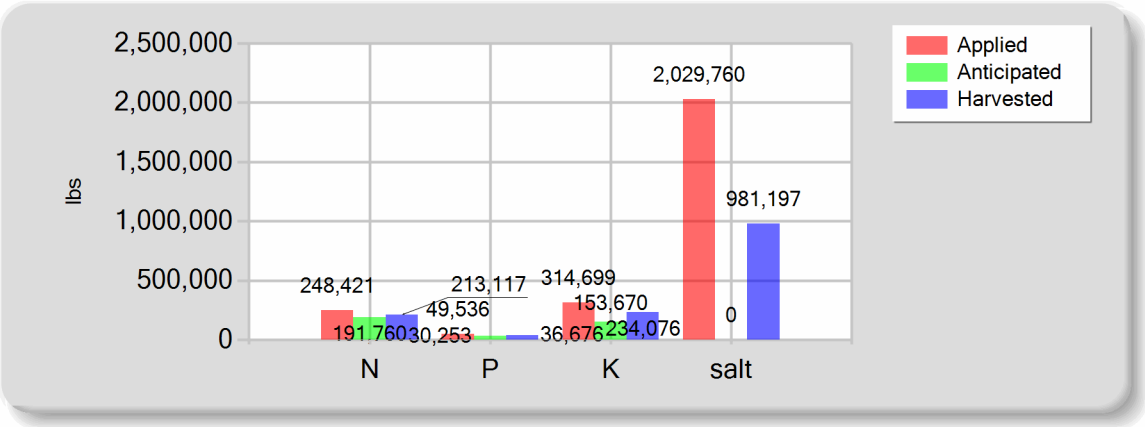
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

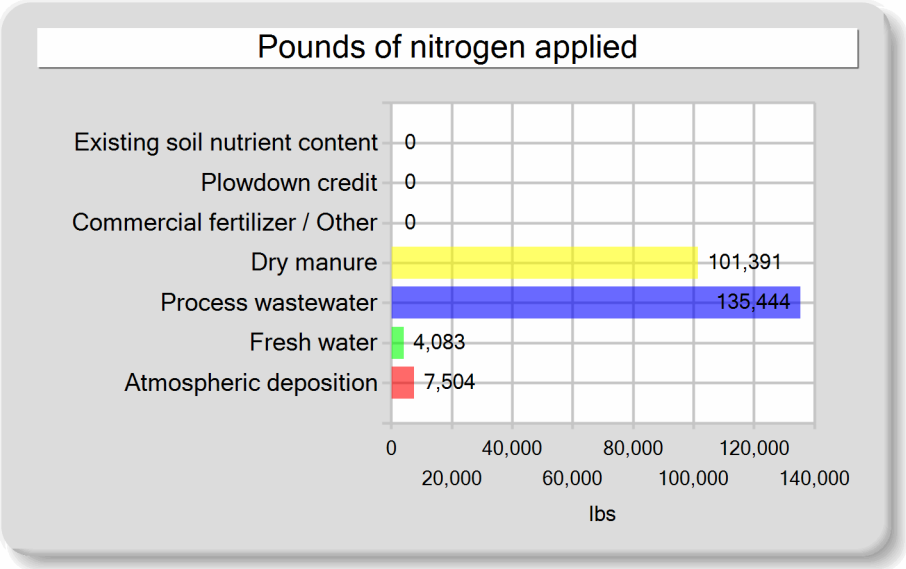
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

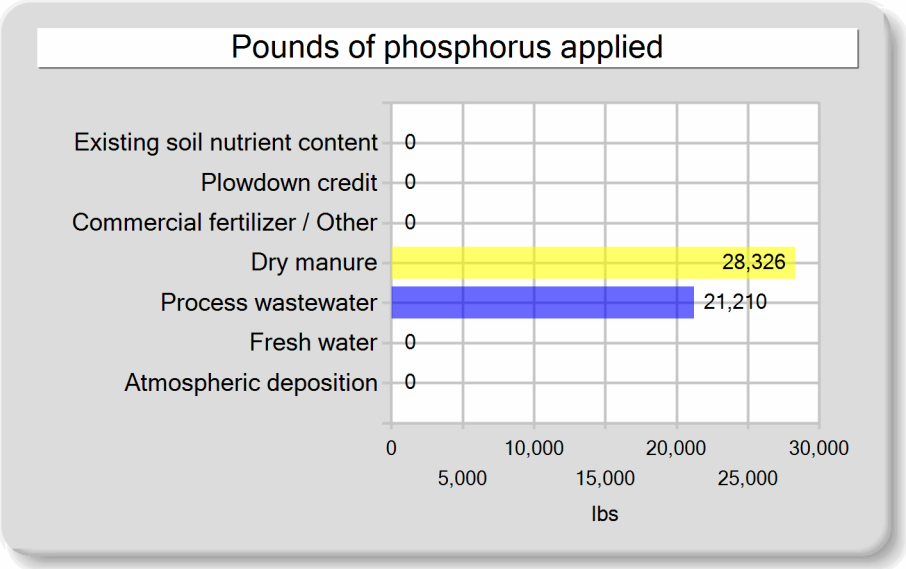
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	101,390.84	28,325.55	131,730.68	0.00
Process wastewater	135,443.81	21,210.11	182,968.48	863,251.28
Fresh water	4,082.78	0.00	0.00	1,166,509.06
Atmospheric deposition	7,504.00	0.00	0.00	0.00
Total nutrients applied	248,421.43	49,535.66	314,699.15	2,029,760.33
Anticipated crop nutrient removal	191,760.00	30,253.00	153,670.00	0.00
Actual crop nutrient removal	213,116.83	36,676.05	234,075.59	981,197.34
Nutrient balance	35,304.61	12,859.61	80,623.56	1,048,562.99
Applied to removed ratio	1.17	1.35	1.34	2.07

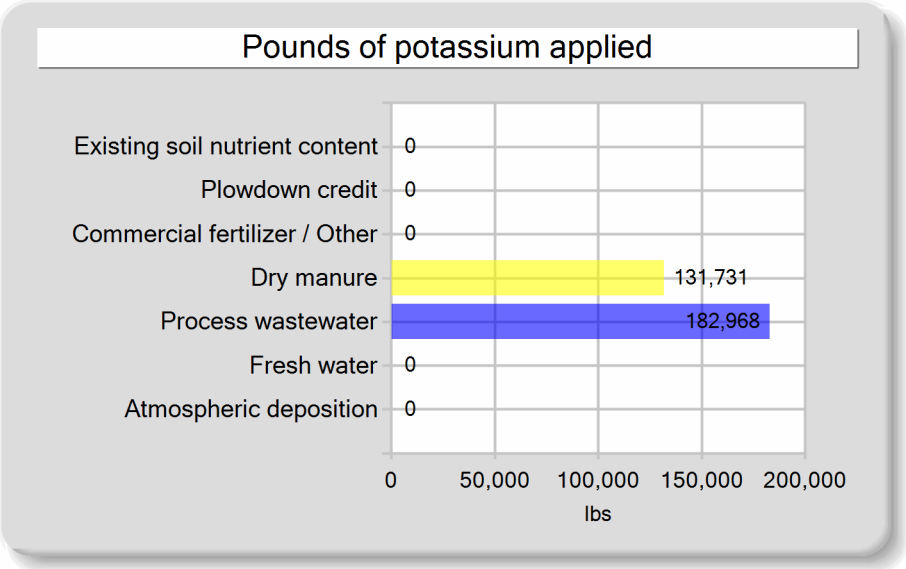
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

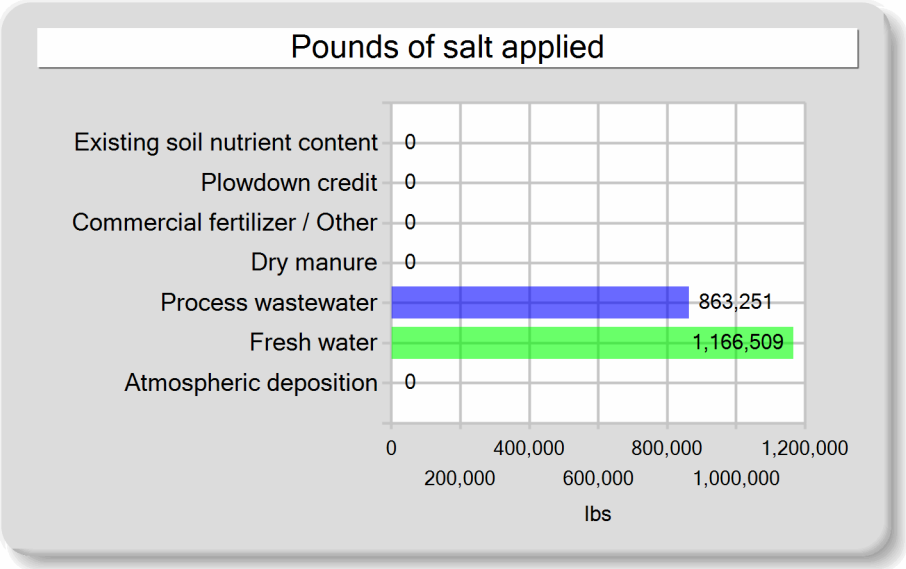


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Louie Nunes

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Louie Nunes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 29, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Lab No.: 23H1168
Sampled By: M. Pedroso
Requested By: Louie Nunes
Submitted Date: 08/10/23
Reported Date: 08/18/23
Project:
Crop ID:

Luis C Nunes Dairy #2
29191 Husman Road
Gustine CA 95322
00-0021363 08

E-mail:
Copy To: cas.labs@yahoo.com

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	TKN mg/L	Total Nitrogen TN mg/L	pH at 25°C unit
1	IW	8/10/23	10:45	1290	1.29	7.6	ND		ND	7.58	7.5

***** = Field NH4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Luis C Nunes Dairy #2
29191 Husman Road
Gustine CA 95322
00-0021363 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23H1168
Sampled By: M. Pedroso
Requested By: Louie Nunes
Submitted Date: 08/10/23
Reported Date: 08/18/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs *calculated from nitrate-N* lbs/AcFoot	lbs/AcInch
1	IW	8/10/23	10:45	**1290**	**1.29**	**7.6**	20.75	1.73

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.

Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).

Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.

MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)

MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.

SM = Standard Methods for the Examination of Water and Wastewater

EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Luis C Nunes Dairy #2
29191 Husman Road
Gustine, CA 95322

Account# 00-0021363
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 08/10/2023 15:29
Reported: 08/18/2023 15:24

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23H1168-01	IW	Ag Water	M. Pedroso		08/10/2023 10:45

Default Cooler Temperature on Receipt °C: -2.9
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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LABORATORY INC

Luis C Nunes Dairy #2
29191 Husman Road
Gustine, CA 95322

Account# 00-0021363
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 08/10/2023 15:29
Reported: 08/18/2023 15:24

Sample Results

Sample: IW
23H1168-01 (Water)

Sampled: 8/10/2023 10:45
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.29**	mmhos/cm	0.01	1		08/14/23 17:11	SM 2510 B		BEH0591
Electrical Conductivity umhos	**1290**	umhos/cm	10.0	1		08/14/23 17:11	SM 2510 B		BEH0591
Ammonia (as N)	ND	mg/L	0.00	1		08/10/23 10:45	Field		BEH0970
Nitrate Nitrogen as NO3N	**7.6**	mg/L	0.1	1	10	08/11/23 16:53	EPA 300.0		BEH0618
pH	**7.5**	units	1.0	1		08/14/23 17:11	SM 4500-H+	H	BEH0591
Temperature	**25.0**	°C	0.0	1		08/14/23 17:11	SM 2510 B		BEH0591
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		08/17/23 09:54	SM 4500-NH3 C		BEH0783
Total Nitrogen	**7.58**	mg/L	1.00	1		08/17/23 09:54	SM 4500-NH3 C		BEH0783

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DELLAVALLE™ LABORATORY INC

Luis C Nunes Dairy #2
29191 Husman Road
Gustine, CA 95322

Account# 00-0021363
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 08/10/2023 15:29
Reported: 08/18/2023 15:24

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0591										
Blank (BEH0591-BLK1)					Prepared: 8/10/2023	Analyzed: 8/14/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	6.0		1.0	units						
Temperature	25.0		0.0	°C						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEH0591-BLK2)					Prepared: 8/10/2023	Analyzed: 8/14/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Blank (BEH0591-BLK3)					Prepared: 8/10/2023	Analyzed: 8/14/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Temperature	25.0		0.0	°C						
Duplicate (BEH0591-DUP1)		**Source: 23H1154-02**			Prepared: 8/10/2023	Analyzed: 8/14/2023				
pH	7.7		1.0	units		7.8			2.19	10
Electrical Conductivity	0.04		0.01	mmhos/cm		0.03			7.39	10
Electrical Conductivity umhos	35.1		10.0	umhos/cm		32.6			7.39	10
Duplicate (BEH0591-DUP2)		**Source: 23H1168-01**			Prepared: 8/10/2023	Analyzed: 8/14/2023				
Electrical Conductivity	1.28		0.01	mmhos/cm		1.29			0.810	10
pH	7.6		1.0	units		7.5			0.531	10
Electrical Conductivity umhos	1280		10.0	umhos/cm		1290			0.810	10
Reference (BEH0591-SRM1)					Prepared: 8/10/2023	Analyzed: 8/14/2023				
Electrical Conductivity	547			umhos/cm	538.0		102	90-110		
Reference (BEH0591-SRM2)					Prepared: 8/10/2023	Analyzed: 8/14/2023				
pH	5.8			units	5.820		99.8	28178-101.71		
Reference (BEH0591-SRM3)					Prepared: 8/10/2023	Analyzed: 8/14/2023				
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEH0591-SRM4)					Prepared: 8/10/2023	Analyzed: 8/14/2023				
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEH0591-SRM5)					Prepared: 8/10/2023	Analyzed: 8/14/2023				
pH	6.0		1.0	units				97.5-102.5		

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DELLAVALLE™
LABORATORY INC

Luis C Nunes Dairy #2
29191 Husman Road
Gustine, CA 95322

Account# 00-0021363
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 08/10/2023 15:29
Reported: 08/18/2023 15:24

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0591 (Continued)										
Reference (BEH0591-SRM6)					Prepared: 8/10/2023	Analyzed: 8/14/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEH0591-SRM7)					Prepared: 8/10/2023	Analyzed: 8/14/2023				
pH	4.0			units	4.000		99.8	97.5-102.5		
Reference (BEH0591-SRM8)					Prepared: 8/10/2023	Analyzed: 8/14/2023				
pH	4.0			units	4.000		99.5	28178-101.7		

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DELLAVALLE™ LABORATORY INC

Luis C Nunes Dairy #2
29191 Husman Road
Gustine, CA 95322

Account# 00-0021363
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 08/10/2023 15:29
Reported: 08/18/2023 15:24

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0618										
Blank (BEH0618-BLK1)					Prepared & Analyzed: 8/11/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0618-BLK2)					Prepared & Analyzed: 8/11/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEH0618-BLK3)					Prepared: 8/11/2023 Analyzed: 8/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEH0618-BS1)					Prepared & Analyzed: 8/11/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		101	90-110		
LCS (BEH0618-BS2)					Prepared: 8/11/2023 Analyzed: 8/12/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		101	90-110		
Duplicate (BEH0618-DUP1)		**Source: 23H1157-01**			Prepared & Analyzed: 8/11/2023					
Nitrate Nitrogen as NO3N	2.0		0.1	mg/L		2.0			0.00	10
Duplicate (BEH0618-DUP2)		**Source: 23H1157-02**			Prepared: 8/11/2023 Analyzed: 8/12/2023					
Nitrate Nitrogen as NO3N	1.6		0.1	mg/L		1.5			0.258	10
Matrix Spike (BEH0618-MS1)		**Source: 23H1157-01**			Prepared & Analyzed: 8/11/2023					
Nitrate Nitrogen as NO3N	7.1		0.1	mg/L	5.000	2.0	101	90-110		
Matrix Spike (BEH0618-MS2)		**Source: 23H1157-02**			Prepared: 8/11/2023 Analyzed: 8/12/2023					
Nitrate Nitrogen as NO3N	6.6		0.1	mg/L	5.000	1.5	101	90-110		
Reference (BEH0618-SRM1)					Prepared & Analyzed: 8/11/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEH0618-SRM2)					Prepared & Analyzed: 8/11/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEH0618-SRM3)					Prepared: 8/11/2023 Analyzed: 8/12/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		

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DELLAVALLE™
LABORATORY INC

Luis C Nunes Dairy #2
29191 Husman Road
Gustine, CA 95322

Account# 00-0021363
Account Manager: Ben Nydam
Submitted By: Louie Nunes

Received: 08/10/2023 15:29
Reported: 08/18/2023 15:24

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEH0783										
Blank (BEH0783-BLK1)					Prepared: 8/16/2023	Analyzed: 8/17/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
Blank (BEH0783-BLK2)					Prepared: 8/16/2023	Analyzed: 8/17/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.00	mg/L						
Total Nitrogen	ND		1.00	mg/L						
LCS (BEH0783-BS1)					Prepared: 8/16/2023	Analyzed: 8/17/2023				
Kjeldahl Nitrogen (TKN), Total	5.83		1.00	mg/L	5.709		102	90-110		
LCS (BEH0783-BS2)					Prepared: 8/16/2023	Analyzed: 8/17/2023				
Kjeldahl Nitrogen (TKN), Total	5.86		1.00	mg/L	5.709		103	90-110		
Duplicate (BEH0783-DUP1)		**Source: 23H1164-04**			Prepared: 8/16/2023	Analyzed: 8/17/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Duplicate (BEH0783-DUP2)		**Source: 23H1364-05**			Prepared: 8/16/2023	Analyzed: 8/17/2023				
Kjeldahl Nitrogen (TKN), Total	ND		1.40	mg/L		ND				10
Matrix Spike (BEH0783-MS1)		**Source: 23H1164-04**			Prepared: 8/16/2023	Analyzed: 8/17/2023				
Kjeldahl Nitrogen (TKN), Total	6.68		1.40	mg/L	7.992	ND	83.6	90-110		
Matrix Spike (BEH0783-MS2)		**Source: 23H1364-05**			Prepared: 8/16/2023	Analyzed: 8/17/2023				
Kjeldahl Nitrogen (TKN), Total	7.77		1.40	mg/L	7.992	ND	97.3	90-110		
Reference (BEH0783-SRM1)					Prepared: 8/16/2023	Analyzed: 8/17/2023				
Kjeldahl Nitrogen (TKN), Total	23.4			mg/L	23.80		98.1	90-110		

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08/10/23 15:29 23H1168

11168

SG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 21363 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Nunes Dairy #2, Luis C
Address: 29191 Husman Road
City, State, Zip: Gustine, CA 95322
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Louie Nunes - (209) 652-7647

Ranch _____

Date sampled: 8/10/23

Sampled by: MPedroso

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples _____ No. Bottles _____

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test**) + TN
(1) 1 L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) 1 L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) 1 L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. 1W	Sampled From:	8/10/23	1045A	Ø	-2.9
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 9/26/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MPedroso	CASINC	8/10/23 1045A	8/10/23 230p
Second				
Third				
Fourth	JV2	DL2	8/10/23 15:29	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____ Date ____

Shipping
$ _____ In
$ _____ Out

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Paste Label Here

08/10/23 15:29 23H1168

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

☐ Samples refridgerated before pick up ☐ Picked up samples placed in Ice chest

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☒ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic	1									
	* pH Value	2									
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1									
	1 L unpreserved (BOD) (Purple) Plastic										
Special	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL VOA, $Na_2S_2O_3$ + MCAA (EPA531)										
	40 mL VOA, $Na_2S_2O_3$ (EPA547)										
	40mL AG VOA unpreserved (White) (Set of 3)										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

pH Strips
Lot: 10BDH4501 Exp: Jan 2025

DRAFT
SEMI-DOUT

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Luis C Nunes Dairy #2
29191 Husman Road
Gustine CA 95322
00-0021363 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0992
Sampled By: M Dekeyrel
Requested By:
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - MB	12/13/23	12:40	1360	1.36	9.2	ND		7.4

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Luis C Nunes Dairy #2
29191 Husman Road
Gustine CA 95322
00-0021363 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0992
Sampled By: M Dekeyrel
Requested By:
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - MB	12/13/23	12:40	**1360**	**1.36**	**9.2**	25.12	2.09

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Luis C Nunes Dairy #2
29191 Husman Road
Gustine, CA 95322

Account# 00-0021363
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:08

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0992-01	DW - MB	Ag Water	M Dekeyrel		12/13/2023 12:40

Default Cooler Temperature on Receipt °C: 4.2
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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LABORATORY INC

Luis C Nunes Dairy #2
29191 Husman Road
Gustine, CA 95322

Account# 00-0021363
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:08

Sample Results

Sample: DW - MB
23L0992-01 (Water)

Sampled: 12/13/2023 12:40
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.36**	mmhos/cm	0.01	1		12/18/23 17:48	SM 2510 B		BEL0704
Electrical Conductivity umhos	**1360**	umhos/cm	10.0	1		12/18/23 17:48	SM 2510 B		BEL0704
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 12:40	Field		BEL0632
Nitrate Nitrogen as NO3N	**9.2**	mg/L	0.1	1	10	12/17/23 07:38	EPA 300.0		BEL0665
Temperature	**25.0**	units	0.0	1		12/18/23 17:48	SM 4500-H+	H	BEL0704
pH	**7.4**	units	1.0	1		12/18/23 17:48	SM 4500-H+	H	BEL0704

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Gustine, CA 95322

Account# 00-0021363
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:08

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0665										
Blank (BEL0665-BLK1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK2)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0665-BLK5)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0665-BS1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.2	90-110		
LCS (BEL0665-BS2)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.8	90-110		
LCS (BEL0665-BS3)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		98.0	90-110		
LCS (BEL0665-BS4)					Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.3	90-110		
Duplicate (BEL0665-DUP1)		**Source: 23L0964-06**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	5.8		0.1	mg/L		5.8			0.639	10
Duplicate (BEL0665-DUP2)		**Source: 23L0989-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	1.3		0.1	mg/L		1.3			0.787	10
Duplicate (BEL0665-DUP3)		**Source: 23L1000-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	0.2		0.1	mg/L		0.2			0.612	10
Duplicate (BEL0665-DUP4)		**Source: 23L1005-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	3.7		0.1	mg/L		3.7			0.941	10
Matrix Spike (BEL0665-MS1)		**Source: 23L0964-06**			Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	10.8		0.1	mg/L	5.000	5.8	102	90-110		
Matrix Spike (BEL0665-MS2)		**Source: 23L0989-02**			Prepared: 12/16/2023 Analyzed: 12/17/2023					
Nitrate Nitrogen as NO3N	6.4		0.1	mg/L	5.000	1.3	102	90-110		

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DELLAVALLE™ LABORATORY INC

Luis C Nunes Dairy #2
29191 Husman Road
Gustine, CA 95322

Account# 00-0021363
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:08

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0665 (Continued)										
Matrix Spike (BEL0665-MS3)			**Source: 23L1000-02**		Prepared: 12/16/2023	Analyzed: 12/17/2023				
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.2	91.6	90-110		
Matrix Spike (BEL0665-MS4)			**Source: 23L1005-02**		Prepared: 12/16/2023	Analyzed: 12/17/2023				
Nitrate Nitrogen as NO3N	8.7		0.1	mg/L	5.000	3.7	99.3	90-110		
Reference (BEL0665-SRM1)					Prepared & Analyzed: 12/16/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0665-SRM2)					Prepared: 12/16/2023	Analyzed: 12/17/2023				
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0665-SRM3)					Prepared: 12/16/2023	Analyzed: 12/17/2023				
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0665-SRM4)					Prepared: 12/16/2023	Analyzed: 12/17/2023				
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0665-SRM5)					Prepared: 12/16/2023	Analyzed: 12/17/2023				
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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DELLAVALLE™
LABORATORY INC

Luis C Nunes Dairy #2
29191 Husman Road
Gustine, CA 95322

Account# 00-0021363
Account Manager: Ben Nydam
Submitted By:

Received: 12/14/2023 14:20
Reported: 12/21/2023 11:08

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0704										
Blank (BEL0704-BLK1)					Prepared & Analyzed: 12/18/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEL0704-BLK2)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEL0704-BLK3)					Prepared & Analyzed: 12/18/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.7		1.0	units						
Duplicate (BEL0704-DUP1)		**Source: 23L0999-01**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.01		0.01	mmhos/cm		1.01			0.248	10
pH	7.7		1.0	units		7.6			0.784	10
Electrical Conductivity umhos	1010		10.0	umhos/cm		1010			0.248	10
Duplicate (BEL0704-DUP2)		**Source: 23L1004-01**			Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1.11		0.01	mmhos/cm		1.11			0.108	10
Electrical Conductivity umhos	1110		10.0	umhos/cm		1110			0.108	10
pH	7.8		1.0	units		7.7			0.775	10
Reference (BEL0704-SRM1)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	426			umhos/cm	426.0		99.9	90-110		
Reference (BEL0704-SRM2)					Prepared & Analyzed: 12/18/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0704-SRM3)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM4)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity	1010			umhos/cm	1000		101	90-110		

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Received: 12/14/2023 14:20
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0704 (Continued)										
Reference (BEL0704-SRM5)					Prepared & Analyzed: 12/18/2023					
Electrical Conductivity umhos	1010			umhos/cm	1000		101	90-110		
Reference (BEL0704-SRM6)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0704-SRM7)					Prepared & Analyzed: 12/18/2023					
pH	4.0			units	4.000		101	97.5-102.5		

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12/14/23 14:20 23L0992

MMM

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 21363 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Nunes Dairy #2, Luis C
Address: 29191 Husman Road
City, State, Zip: Gustine, CA 95322
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Louie Nunes - (209) 652-7647

Ranch _____

Date sampled: 12/13/23

Sampled by: M. Dekeyrel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples _____ No. Bottles _____

Water Type: [ ] Drinking [ ] Wastewater
[ ] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(✓) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-MB	Sampled From:	12/13	12:40pm	0	4.2
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M Dekeyrel	CAS INC	12/13/23 @ 12:40pm	12/14/23 12p
Second	JLF	DLV	12-14-23 12:00	12-14-23 2:15
Third				
Fourth		DLV	12/14 14:20	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

12/14/23 14:20 23L0992

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1									
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT